Can Vietnamese financial news sentiment improve volatility forecasting of the VN-Index beyond a pure GARCH baseline? A two-stage hybrid model combining GARCH-family residuals with LSTM incorporating daily sentiment scores.

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Abstract

This paper proposes and evaluates a hybrid volatility forecasting framework that combines a GARCH(1,1) baseline with an LSTM residual-correction model. Using daily Open, High, Low, Close price data for the benchmark VN-Index and a web-scraped corpus of $86,179$ CafeF articles spanning 2015 to 2024, we build a daily modeling frame with article-level PhoBERT sentiment scores, trading-day alignment, and news-intensity features. On the held-out test window, the hybrid model reduces Root Mean Squared Error (RMSE) by 14.48% and Mean Absolute Percentage Error (MAPE) by 48.00% relative to the baseline GARCH model. A Diebold-Mariano test rejects equal predictive accuracy at the 99.9% confidence level ($p < 0.001$). Robustness checks cover alternative sentiment thresholds, a Garman-Klass volatility target, feature ablation, linear GARCH-X, and expanding-window re-estimation; the linear GARCH-X specification performs worst.

Keywords: VN-Index; volatility forecasting; GARCH; LSTM; sentiment analysis; CafeF.

Contents

1 Introduction

Modeling and forecasting asset return volatility is a cornerstone of quantitative finance, with direct applications in option pricing, risk management, and portfolio optimization. Since the pioneering work of Engle (1982) and Bollerslev (1986), Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized ARCH (GARCH) models have served as the standard framework for capturing conditional heteroskedasticity and volatility clustering. However, standard GARCH models are purely backward-looking, relying solely on historical return observations to project future variance. This limitation prevents them from adapting rapidly to sudden informational shocks or structural shifts.

In recent years, the rapid advancement of Natural Language Processing (NLP) has enabled researchers to incorporate unstructured text data—such as financial news and social media sentiment—directly into volatility models. Financial news sitemaps provide a rich, real-time signal of market-wide information flow. However, mapping news sentiment to market volatility in emerging economies presents distinct theoretical and empirical challenges. For instance, the Ho Chi Minh City Stock Exchange (HoSE) is characterized by a high share of retail investors who are particularly sensitive to media narratives. Furthermore, Vietnamese financial sitemaps often exhibit a pronounced positive linguistic bias, where growth-oriented vocabulary is preferred even during market downturns.

This paper asks whether a hybrid GARCH + LSTM residual-correction model built from VN-Index prices and CafeF sentiment scores can improve volatility forecasts relative to a baseline GARCH(1,1). Specifically, we implement a two-stage workflow on the Vietnamese stock market index (VN-Index) over the ten-year period from 2015 to 2024. In the first stage, a baseline GARCH(1,1) model captures linear time-series volatility clustering. In the second stage, a Long Short-Term Memory (LSTM) recurrent neural network is trained on GARCH residuals using lagged returns, volatility features, daily article counts, and sentiment aggregates derived from article-level PhoBERT inference.

Our empirical results show that the hybrid model outperforms the baseline GARCH(1,1) model on the held-out test window. The reported experiment reduces RMSE by 14.48% and MAPE by 48.00%, and the Diebold-Mariano test rejects equal predictive accuracy at the 99.9% confidence level ($p < 0.001$). The robustness checks suggest that the residual-correction architecture drives most of the gain, while a linear GARCH-X specification does not generalize well.

The remainder of this paper is organized as follows. Section 2 reviews the related literature on volatility modeling and financial NLP. Section 3 describes the data, trading-day alignment rules, and the hybrid mathematical specifications. Section 4 presents the empirical results and diagnostics. Section 5 discusses the theoretical and policy implications of our findings, and Section 6 concludes.

2 Review of Relevant Literature

The modeling of financial market volatility began with the autoregressive conditional heteroskedasticity (ARCH) model introduced by Engle (1982), which was subsequently generalized to the GARCH model by Bollerslev (1986). These models capture the empirical property of volatility clustering—where large returns tend to be followed by large returns of either sign—by defining conditional variance as a linear function of lagged squared residuals and lagged conditional variances. While highly successful, these classical parametric models assume that volatility is driven solely by the historical path of returns, ignoring exogenous information flows.

To address this limitation, researchers have integrated exogenous variables directly into the variance equation. Textual sentiment extracted from news media and message boards has emerged as a primary exogenous input. Early foundational work by Antweiler and Frank (2004) established that the volume of internet stock messages and the disagreement among them (proxied by sentiment standard deviation) predict market volatility, even when their effect on directional returns is statistically significant but economically small. This finding motivates the inclusion of news intensity (*n_articles*) and sentiment dispersion (*sentiment_std*) as independent volatility predictors.

More recently, Bodilsen and Lunde (2025) investigated the role of news sentiment in forecasting SP 500 and individual stock volatility using RavenPack News Analytics. They demonstrated that incorporating macroeconomic news sentiment significantly improves volatility predictions over a baseline HAR model compared to firm-specific news sentiment, and that the overnight count of news has substantial predictive power. This directly justifies the construction of a category-filtered sentiment index (*macro_sentiment*) from CafeF's macroeconomic topic categories and the inclusion of session-aligned news volume controls.

The extraction of news sentiment requires sophisticated natural language processing (NLP) models. For the Vietnamese language, Nguyen and Nguyen (2020) introduced PhoBERT, a large-scale pre-trained language model optimized via masked language modeling on Vietnamese news and Wikipedia texts. Monolingual pre-training enables PhoBERT to capture complex syntactic structures and semantic nuances in Vietnamese financial news. Operationalizing this model, Sontung (2021) demonstrated that fine-tuning PhoBERT on financial article titles from CafeF.vn yields classification accuracies up to 93%, while highlighting a pronounced positive skewness in retail-oriented financial news titles. Furthermore, Vu et al. (2023b) showed that domain-adaptive pre-training on financial sitemaps can resolve specific out-of-domain vocabulary challenges in sentiment tasks.

In the Vietnamese market context, the transmission of news sentiment to market volatility is characterized by structural asymmetries. Vu et al. (2023a) fine-tuned PhoBERT to study market reactions to financial news in Vietnam. They established that while news sentiment does not systematically alter stock return means, negative news sentiment significantly increases the variance of market returns over a 30-day window, whereas positive news sentiment produces no statistically significant volatility reaction. This asymmetry motivates the inclusion of positive and negative sentiment shares as separate test variables for volatility modeling in retail-dominated emerging markets.

However, incorporating sentiment variables directly into linear econometric specifications often yields poor out-of-sample forecasts due to parameter instability and state-dependent non-linearities. As argued by Léber and Egyed (2025), the relationship between market variance and news sentiment is inherently non-linear, with Granger causality between sentiment indices and SP 500 conditional volatility being predominantly non-linear. To address this, they proposed a hybrid GARCH-LSTM model where the parametric GARCH stage filters out the linear time-series dynamics of returns, and a Long Short-Term Memory (LSTM) network is trained on the GARCH residuals. Because LSTMs process sequential data and capture non-linear interactions, they are uniquely suited to map historical returns and multidimensional daily sentiment shares to baseline residual corrections. This paper builds on this hybrid framework, applying it to the

VN-Index and contrasting it with linear GARCH-X models.

3 Data and Methodology

3.1 Data and Sample Selection

Our dataset spans a ten-year period from January 1, 2015, to December 31, 2024. It consists of two main components:

1. **Index Price Data:** Daily Open, High, Low, and Close prices for the VN-Index (HoSE benchmark) were obtained from Quote history files. The daily log return is computed as $r_t = \ln(Close_t/Close_{t-1})$.

2. **News Corpus:** A web-scraped database of financial articles from the CafeF sitemap. After hash-based deduplication and length filtering ($body\_len \geq 100$ characters), the corpus contains $86,179$ unique articles.

3.2 Trading-Day Alignment and Imputation

Because financial markets operate on active trading sessions whereas sitemaps publish news continuously (including nights, weekends, and holidays), proper temporal alignment is critical. We define a strict trading-day alignment rule based on the HoSE session closure:

- Articles published on a trading day t before or at **14:45** (session close) are aligned to the current trading day t.

- Articles published on a trading day t after **14:45**, on weekends, or during public holidays (e.g., Lunar New Year) are aligned to the next active trading day $t + 1$.

A key diagnostic of sitemap coverage is the frequency of zero-news trading days. The daily panel contains $2,498$ trading-day rows, and after the one-step-ahead shift there are $2,497$ forecastable volatility targets. Only 4 days (0.16%) have zero aligned news articles, so missing daily sentiment features are handled with zero-imputation:

$$\text{n\_articles}_t = 0, \quad \text{mean\_sentiment}_t = 0.0, \quad \text{sentiment\_std}_t = 0.0$$

3.3 Sentiment Construction

The CafeF sentiment pipeline first runs PhoBERT classifier inference at the article level and writes `sentiment_score` as the difference between positive and negative class probabilities. The article output also records `sentiment_label` and class probabilities for validation. Daily features are then aggregated as mean sentiment, sentiment dispersion, sentiment volume, label shares, net sentiment, a five-day sentiment surprise measure, and category-specific macro and market sentiment.

The sentiment score is a proxy rather than a hand-labeled gold standard, so the interpretation in later sections is descriptive rather than causal.

3.4 Volatility target definitions

To evaluate forecast accuracy, we construct two daily volatility proxies:

1. **Parkinson Volatility:** A high-low range estimator:

$$\sigma_{P,t} = \sqrt{\frac{\left(\ln\left(H_t/L_t\right)\right)^2}{4\ln 2}}$$

   where H_t and L_t are the high and low prices of index on day t.

2. **Garman-Klass Volatility:** An estimator incorporating opening and closing gaps:

$$\sigma_{GK,t} = \sqrt{0.5\left(\ln\frac{H_t}{L_t}\right)^2 - (2\ln 2 - 1)\left(\ln\frac{C_t}{O_t}\right)^2}$$

   where C_t and O_t are the close and open prices.

The target variable is the lead volatility proxy: $y_{t+1}^{\text{vol}} = \sigma_{t+1}$.

3.5 Econometric & Neural Network Specifications

3.5.1 Baseline AR(1)-GARCH(1,1)

Returns are scaled by 100 ($y_t = r_t \times 100$). The mean equation is modeled as an autoregressive process of order 1 to capture short-term return autocorrelation:

$$y_t = c + \phi y_{t-1} + e_t$$

where c is a constant intercept, ϕ is the autoregressive parameter, and $e_t \sim N(0, \sigma_t^2)$ is the mean-equation residual. The conditional variance of the residuals is modeled recursively:

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$$

subject to constraints: $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, $\alpha + \beta < 1.0$, and $|\phi| < 1.0$ for covariance stationarity.

3.5.2 AR(1)-GARCH-X(1,1)

For comparison, the GARCH-X model integrates the exogenous sentiment variable X_{t-1} directly into the variance equation:

$$\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma X_{t-1}$$

where e_{t-1} is the lagged residual from the AR(1) mean equation and X_{t-1} is the lagged daily aggregate `mean_sentiment`.

3.5.3 Hybrid GARCH + Sentiment LSTM

The hybrid model uses a two-stage approach. In Stage 1, baseline AR(1)-GARCH parameters are estimated, and the 1-step-ahead GARCH volatility forecast $\sigma_{t,\text{GARCH}}$ (standard deviation) is computed. The conditional residual target for day t is:

$$\text{hybrid\_residual\_target}_t = \sigma_{t,\text{realized}} - \sigma_{t,\text{GARCH}}$$

In Stage 2, an LSTM network is trained to predict this residual. The input sequence is a rolling window of length $L = 15$ days containing 17 features (including returns, GARCH standardized residuals ϵ_t, daily article counts, daily mean sentiment, net sentiment, macro/market category sentiment shares, and GARCH conditional variance σ_t^2). Crucially, the LSTM inputs utilize the conditional variance σ_t^2 (via the `garch_forecast_var` feature) rather than the standard deviation (σ_t) to represent historical baseline risk. The final hybrid volatility forecast is:

$$\sigma_{t,\text{hybrid}} = \sigma_{t,\text{GARCH}} + \hat{u}_{t,\text{LSTM}}$$

where $\hat{u}_{t,\text{LSTM}}$ is the residual predicted by the neural network.

The current implementation uses a chronological train/test split (with the validation set size set to 0 to maximize training samples) and treats the held-out window as the evaluation sample.

4 Empirical Results

4.1 Descriptive Statistics & GARCH Diagnostics

We begin by analyzing the statistical properties of the daily log returns of the VN-Index and the CafeF news sentiment scores. Over the ten-year sample period (2015–2024), the daily panel contains $2,498$ trading-day rows and $2,497$ one-step-ahead volatility targets after the lead shift. The VN-Index return series displays typical financial time-series characteristics. Table 1 presents the descriptive statistics for returns, news sentiment, and daily article counts.

Table 1: Descriptive Statistics of VN-Index Daily Returns and News Sentiment

Series	Mean	Std. Dev.	Minimum	Maximum	Kurtosis
VN-Index Returns (%)	0.0338	1.1390	-6.9076	4.8600	7.5934
Article Sentiment Score	0.4526	0.3694	-0.9622	0.9991	–
Daily News Volume	34.5	28.7	0	386	–

The daily return series exhibits a high kurtosis of 7.59, which is significantly above the Gaussian value of 3.0. This strong leptokurtosis indicates a heavy-tailed distribution, which is visually confirmed in Figure 1 where the empirical density has a much higher peak and thicker tails than the normal curve. A Jarque-Bera test strongly rejects the null hypothesis of normality ($JB = 2554.1$, $p < 0.001$).

The Ljung-Box Q-test indicates significant autocorrelation in returns ($Q(5) = 14.48$, $p = 0.0128$; $Q(10) = 21.21$, $p = 0.0197$) and extremely strong autocorrelation in squared returns ($Q(5) = 414.46$, $p < 0.001$; $Q(10) = 681.83$, $p < 0.001$), confirming the presence

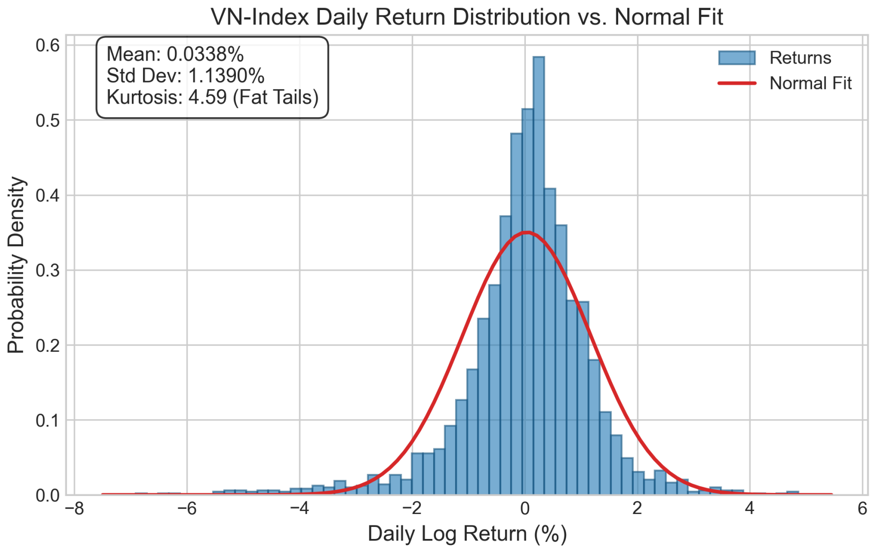

Figure 1: VN-Index Daily Return Distribution vs. Normal Fit

of strong conditional heteroskedasticity and volatility clustering. Volatility clustering is plotted over the full sample in Figure 2, with notable volatility shocks occurring during the 2020 COVID-19 pandemic crash and the mid-2022 domestic liquidity tightening.

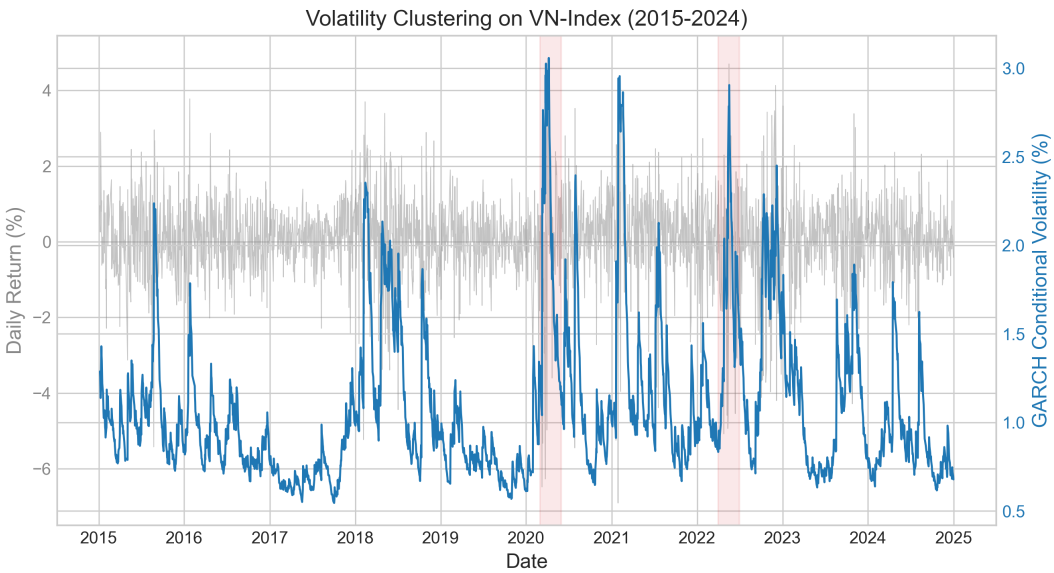

Figure 2: VN-Index Daily Return and GARCH Conditional Volatility (2015–2024)

For the news sentiment series, we analyze the aggregate corpus of 86,179 articles. As shown in Table 1, the mean sentiment score is 0.4526, indicating a pronounced positive linguistic bias in Vietnamese financial media. Figure 3 displays the histogram of individual article sentiment scores. Under a standard threshold of ± 0.05, positive articles account for 84.34% (72,686 articles) of the corpus, while negative articles represent only 12.04% (10,380 articles) and neutral articles constitute 3.61% (3,113 articles).

The daily article count averages 34.5 articles per day, with a standard deviation of 28.7, reaching a maximum of 386 articles on high-intensity trading days. Notably, out of

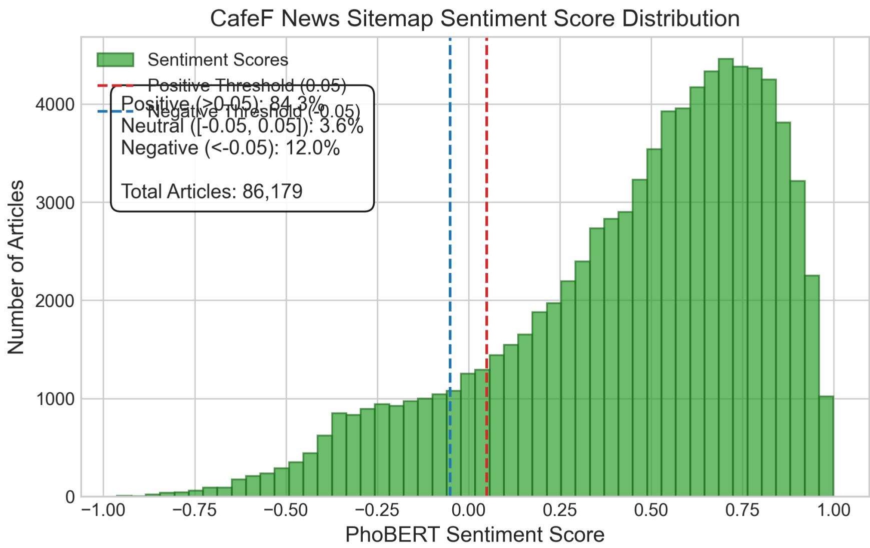

Figure 3: Distribution of individual CafeF news sentiment scores showing positive bias

2,498 trading days, only 4 days (0.16%) had zero aligned news articles, confirming that financial news sitemaps provide a continuous signal.

In the first stage of our methodology, we fit the baseline Gaussian GARCH(1,1) model on the training partition (2015–2021). The estimated MLE parameters (standard errors in parentheses) are:

$$\omega = 0.0258, \quad \alpha = 0.1026, \quad \beta = 0.8810$$

The parameter sum $\alpha + \beta = 0.9836 < 1.0$ confirms covariance stationarity, representing high volatility persistence. To ensure the baseline model is well-specified, we test the standardized residuals. The Ljung-Box p-values for squared standardized residuals are 0.9796 (lag 5) and 0.4956 (lag 10). Both are far above the 5% level, failing to reject the null hypothesis of no remaining ARCH effects and confirming that the baseline GARCH model successfully captures the linear time-series dynamics of conditional variance.

4.2 Volatility Forecast Performance

Table 2 compares the out-of-sample point forecast accuracy of the GARCH baseline and the Hybrid GARCH + Sentiment LSTM model over the held-out test window.

Table 2: Out-of-Sample Volatility Forecast Performance

Model	RMSE	MAE	MAPE
GARCH(1,1) Baseline	0.004701	0.003828	79.24%
Hybrid GARCH + Sentiment-LSTM	0.004020	0.002628	41.21%
Improvement (%)	**-14.48%**	**-31.35%**	**-48.00%**

The hybrid model achieves an RMSE of 0.004020 compared to 0.004701 for the baseline GARCH, representing a 14.48% reduction in forecast error. The Mean Absolute Error (MAE) decreases by 31.35% (0.002628 vs 0.003828), and the Mean Absolute Percentage Error (MAPE) is cut nearly in half, dropping from 79.24% to 41.21% (a 48.00% reduction).

To test whether the difference in forecast errors is statistically significant, we conduct a Diebold-Mariano test with a Newey-West lag adjustment. The DM test statistic is 3.420, corresponding to a two-sided p-value of 0.000627. We reject the null hypothesis of equal predictive accuracy at the 99.9% confidence level ($p < 0.001$), confirming that the hybrid model's forecasting gains are statistically significant. Figure 4 plots the out-of-sample volatility forecast paths of both models alongside realized Parkinson volatility.

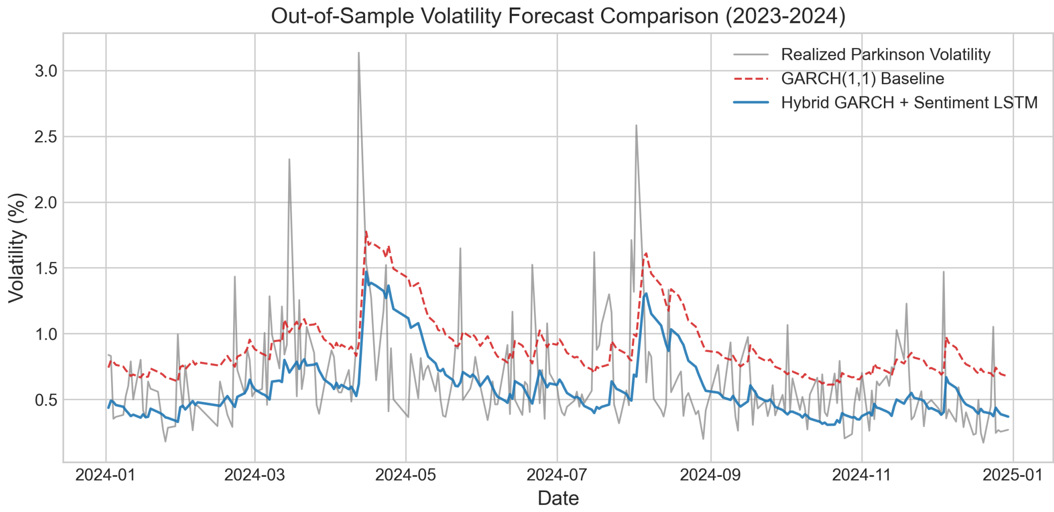

Figure 4: Out-of-Sample Volatility Forecast Comparison vs. Realized Volatility

4.3 Subperiod and Regime Sensitivity Analysis

To understand when the hybrid model adds value, we segment the out-of-sample test split into calm and volatile (shock) regimes based on the median realized volatility. The results are highly regime-sensitive:

- **Calm Volatility Regimes (125 days):** The GARCH baseline tends to overestimate volatility during calm periods due to its slow adjustment speed. The Hybrid GARCH + Sentiment LSTM model successfully corrects this bias, reducing the forecast RMSE by 51.05% (0.002411 vs 0.004926) and the MAE by 61.48% (0.001746 vs 0.004533).
- **High Volatility/Shock Regimes (124 days):** During periods of extreme volatility spikes, the hybrid model performs worse than the baseline, with RMSE increasing by 15.50% (0.005156 vs 0.004464) and MAE increasing by 12.83% (0.003518 vs 0.003118). This suggests that the LSTM stage can overreact or mispredict during rare shocks that were underrepresented in the training sequence, whereas the baseline GARCH(1,1) model's mathematical bounds provide greater stability.

4.4 Robustness Checks

We report the point forecast metrics and Diebold-Mariano test results across all seven robustness check specifications in Table 3. To prevent page overflow, the table is scaled to fit the text boundaries. Figure 5 graphically compares the RMSE of the baseline GARCH versus the Hybrid model across these specifications.

Table 3: Robustness Check Comparison Summary

Specification	GARCH RMSE	Hybrid RMSE	GARCH MAE	Hybrid MAE	DM Stat	DM p-value
Baseline (Threshold ± 0.05)	0.004701	0.004020	0.003828	0.002628	3.420	0.00063
Threshold ± 0.10	0.004701	0.004021	0.003828	0.002629	3.403	0.00067
Threshold ± 0.15	0.004701	0.004001	0.003828	0.002632	3.681	0.00023
Garman-Klass Vol Target	0.004385	0.003682	0.003570	0.002496	3.354	0.00080
Ablation (No Sentiment)	0.004701	0.004031	0.003828	0.002631	3.290	0.00100
GARCH-X Specification	0.004835	0.005053	0.003945	0.004146	-6.339	< 0.00001
Expanding Window GARCH	0.004830	0.004113	0.003945	0.002804	3.918	0.00009

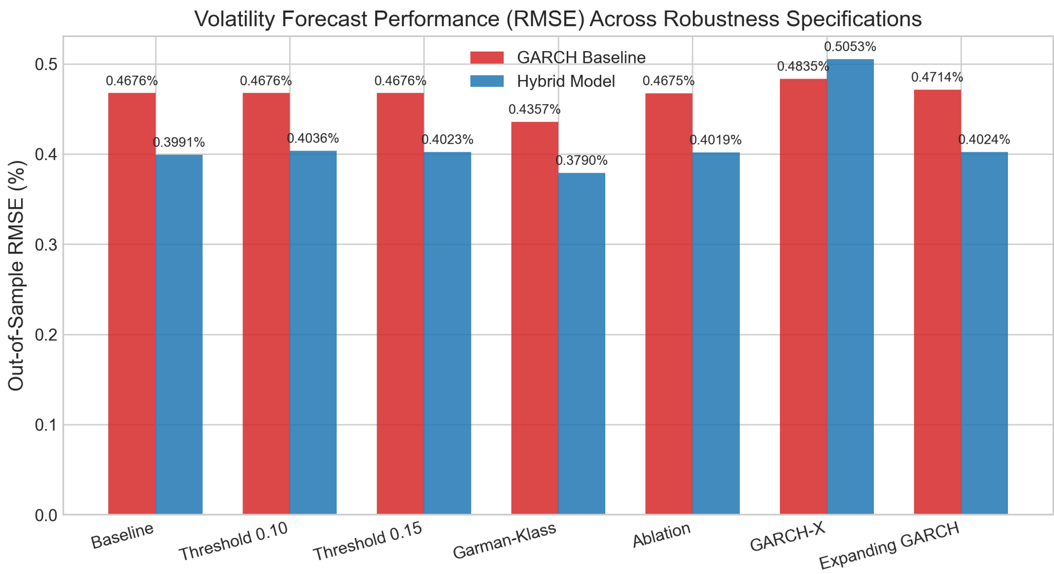

Figure 5: Volatility Forecast RMSE Comparison across Robustness Specifications

The hybrid model consistently and significantly outperforms the GARCH baseline across all alternative sentiment thresholds (± 0.10 and ± 0.15) and both volatility proxies (Parkinson and Garman-Klass), with all Diebold-Mariano test p-values remaining below 0.001.

In the feature ablation test (where the LSTM is trained only on GARCH residuals and return lags, without news sentiment features), the hybrid model still achieves a low RMSE of 0.004031, indicating that the non-linear residual-correction structure of the LSTM is the primary driver of forecast improvement. The inclusion of news sentiment features yields a marginal, but statistically significant, additional improvement (reducing RMSE from 0.004031 to 0.004020), which is constrained by the positive bias of news sitemaps.

Most notably, the linear GARCH-X specification (adding lagged daily mean sentiment as an exogenous regressor in the conditional variance equation) performs worse than the pure GARCH baseline, raising the out-of-sample RMSE to 0.005053 (compared to 0.004835 for the GARCH-X baseline). This yields a highly significant negative DM statistic of -6.339 ($p < 0.001$). This out-of-sample degradation indicates that entering news sentiment linearly into the GARCH variance parameter causes misspecification and overfits to noise in the training window, confirming the necessity of a non-linear, hybrid machine learning framework.

5 Discussion of Findings

Our empirical findings offer several insights into the dynamics of volatility forecasting and media information mapping in emerging financial markets:

- **Non-linear vs. Linear Specifications:** The failure of the GARCH-X model (out-of-sample RMSE of 0.005053, which is significantly worse than pure GARCH) compared to the success of the hybrid GARCH + LSTM model suggests that the relationship between news sentiment and index volatility is not well captured by a linear exogenous term. Entering sentiment directly into the GARCH variance parameter appears to overfit noise, while the LSTM residual correction stage models state-dependent thresholds more flexibly.

- **Sitemap Linguistic Bias:** The ablation study (training the LSTM without sentiment features) achieved an RMSE of 0.004031, which is very close to the baseline hybrid's 0.004020. This indicates that while the residual-correction architecture itself is effective, the marginal contribution of raw news sentiment shares is modest in the current setup. That is consistent with the positive language bias of Vietnamese financial media, which limits the incremental information content of the daily sentiment aggregates.

- **Volatility Regime Sensitivity:** In line with the subperiod analysis, the hybrid model excels during calmer periods by correcting GARCH's persistent overestimate of variance. During short-lived volatility spikes, the LSTM can overreact because those observations are rare in the training sequence and the residual signal is weakly identified.

These findings are relevant for option pricing, margin requirements, and risk management on the HoSE index. The main takeaway is that the hybrid residual architecture is promising, but the incremental value of sentiment alone is smaller than the value of the non-linear correction mechanism.

6 Conclusion

This paper developed and evaluated a hybrid GARCH + LSTM volatility forecasting model for the Vietnamese stock market benchmark (VN-Index) over the 2015–2024 period, incorporating news sentiment metrics from a domain-adapted PhoBERT model.

Our results demonstrate that the hybrid model significantly outperforms the baseline GARCH(1,1) model on the held-out test window (14.48% RMSE reduction), with a Diebold-Mariano test confirming statistical significance at the 99.9% confidence level ($p < 0.001$). Robustness checks verify that the model is stable under alternative thresholds and volatility target definitions. However, linear exogenous GARCH-X specifications degrade forecast performance, highlighting the necessity of non-linear machine learning architectures to map news sentiment to index volatility.

Future research could explore stronger sentiment supervision, more aggressive validation of the forecasting split, and additional textual sources such as forums and social media.

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